

Combining AI and Crowdsourcing to Identify Misinformation in Online News

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Abstract

Misinformation in online news presents a critical challenge as it can influence individuals' assessment of news articles and impact decision-making. Existing approaches to combat misinformation rely on either automated AI-based systems or human-centered methods, such as crowdsourcing. However, AI systems often struggle to interpret subjective opinions and remain up to date, while human approaches can be biased. To address these, we designed and developed AI and People News Evaluation (APNE), a system that combines both AI and human assessment to identify misinformation. An initial evaluation of our system suggests that participants valued AI for offering factual information but sometimes felt it was biased. On the other hand, humans, despite having subjective opinions, shared facts and alternative sources that improved the information quality. We also found that participants' assessments were shaped by external factors including their distrust of AI and their opinions about other readers.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools; Empirical studies in HCI.**

Keywords

News article, Misinformation, AI, Crowdsourcing

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1 Introduction

In recent years, online news sites have become the dominant source of information with 66% of people in the US getting their news from online news sites [3]. While all news sites boast reliability, assessing the trustworthiness of articles published on these sites remains challenging. Even reputable sources often exhibit biases [21] and perpetuate echo chambers [7], while many other sites are intentionally malicious and misleading [16].

This issue is particularly critical because many rely on online news to make important decisions. For instance, during the COVID-19 pandemic, conspiracy theories about vaccines spread widely, posing serious risks to public health [17, 22]. Similarly, during the 2024 U.S. elections, numerous biased articles about polls aimed to sway public opinion in specific directions [20]. These examples underscore the urgent need for tools that can help people navigate the complex landscape of online news and distinguish factual information from misinformation.

Prior works in HCI have proposed several approaches to address the problem of misinformation. Many of these approaches rely on computational methods, such as machine learning algorithms [4] or large language models [26], while others have more human-centered approaches including crowd-sourcing [8]. However, systems that solely rely on human input can be inconsistent, as personal biases often influence assessments [31], and the evaluations on article quality may not always be based on factual evidence [5]. Conversely, AI-based systems, while consistent and scalable, are limited by their inability to make predictions on updated information in real time and challenges in interpreting subjective opinions within news articles [27].

In this work, we take a first step toward building systems that can address misinformation by leveraging both crowdsourcing and AI-based approaches. To that end, we designed and developed AI and People News Evaluation (APNE), a system that combines AI's objectivity alongside humans' nuanced understanding of subjectivity. This hybrid approach aims to help users make more informed decisions and construct well-founded opinions about current affairs. APNE consists of three main components. The foundational component is the interactive article that allows readers to comment on individual sentences and images. The two complementary components are the AI analysis and the user-generated input sections. Both of these components enable the assessment of the news article based on four key criteria: accuracy, reliability, unbiasedness, and trustworthiness. Additionally, each component provides a textual and visual summary of ratings based on these criteria, offering readers a comprehensive assessment of the article at a glance. To enhance transparency and build trust, both components also provide explanatory details to justify the rating. To evaluate the effectiveness of APNE, we conducted an initial study with 20 participants. Our findings suggest that while AI can provide factual information, it can also propagate bias, and people can provide facts and alternative sources that could better inform other readers. We also found that participants' opinions on the articles were influenced by factors beyond the article content including source, their mistrust of AI, pre-existing beliefs, and their judgment toward other readers.

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2 Related Works

2.1 Automatic Fake News Detection Using AI

Prior works explored automatic fake news detection through text classification approaches using various machine learning (ML) or deep learning (DL) models [2]. For example, Villela et al. [24] compared different ML algorithms and found that ensemble methods, which combine multiple models, achieved the highest accuracy in identifying fake news. Detection methods that utilize large language models (LLMs), such as GPT [9], or transformer-based systems such as BERT [11], also demonstrate high accuracy. However, these systems are often limited by their training datasets.

As news content evolves rapidly, these models often struggle to stay updated with real-time events. Additionally, many of these models are based primarily on textual data. Although there has been research on multimodal fake news detection [23], these approaches face challenges around inconsistencies in matching images with text, such as outdated or unrelated images used in articles. Another limitation is the lack of interpretability in AI models [1, 25]. Users often find it difficult to trust the outputs of these systems due to insufficient explanations of their decisions.

2.2 Human-Centered Approach to Misinformation Detection

Other researchers explored human-centered approaches to misinformation detection through expert fact-checking [18] or crowdsourcing [10]. Expert-driven methods, such as manual fact-checking, are demonstrated to be highly accurate [19]. However, they are often time-intensive and expensive. Alternatively, crowdsourcing platforms, such as community notes on social media (e.g., X) engage the public in fact-checking. Researchers have also explored approaches that enable others to share their opinions on articles [28]. This approach allows users to comment on subtleties such as tone or context that AI techniques might overlook. However, these methods also have significant drawbacks around biases, as people often struggle to remain objective [5]. Additionally, many crowdsourcing systems require participants to classify content as simply "true" or "false." Such binary approaches are often insufficient for nuanced cases, where the truth may involve multiple aspects or degrees of validity [12].

2.3 Combining AI and Crowdsourcing

Prior works suggest that both approaches have their strengths and weaknesses. While AI-based methods excel in accuracy and factual consistency, they often fail to detect misleading language or adapt to rapidly evolving news. Conversely, human-centered approaches are better at identifying complex relationships between text and images or understanding nuanced contexts. However, human reviewers can introduce bias and are constrained by their prior knowledge. A promising direction is to develop hybrid systems that integrate the strengths of both approaches. Such systems could leverage the factual objectivity and scalability of AI-based methods while incorporating human input for complex or context-dependent cases to build more robust misinformation detection systems.

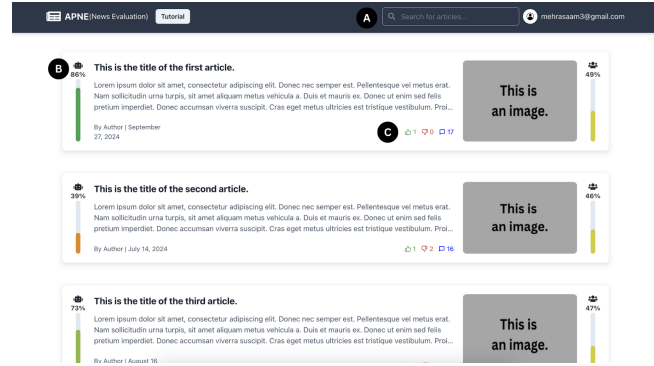


Figure 1: APNE Home Page A) Search option for articles. B) Article card containing title, author's details, publication date, and first paragraphs and two bars showing AI and People score. C) User interactions, number of likes, dislikes, and comments. (Due to copyright restrictions, original images and text have been replaced with placeholders and Lorem Ipsum text.)

3 APNE (AI and People News Evaluation)

We designed and developed APNE as a news-reading system to assist users in making more informed decisions about the reliability of news articles. The system integrates both AI and human assessment to evaluate news content based on four key criteria: accuracy, reliability, trustworthiness, and unbiasedness. To deliver information to readers, APNE uses visualizations, including rating scores, charts, and detailed explanations.

3.1 Home Page

The home page provides a search option in the navigation bar, accompanied by the reader profile and ID (Fig. 1 (A))¹. On this page, readers can find news article cards. Each article card (Fig. 1 (B)) displays a summary of the article's content, including the title, image, author's details, publication date, and the first paragraph of the article.

On either side of each article card, there are two vertical bars with corresponding icons for AI and people. These bars present the article rating scores assigned to the article by both other readers and the AI system. The lines are also color-coded, with colors ranging from red to green: a higher score corresponds to green, while a lower score is indicated by red. Each card also summarizes reader interactions (Fig. 1 (C)), such as the number of likes, dislikes, and comments the article has received. When a reader clicks on an article card, they are directed to the article's detailed page.

3.2 Article Page

This page contains relevant information about the article, organized into three components: AI, Article, and People. Each component provides distinct insights into the article. It is designed to allow readers to distinguish between AI-generated assessments from human

¹Due to copyright restrictions, original images and text have been replaced with placeholders and Lorem Ipsum text.

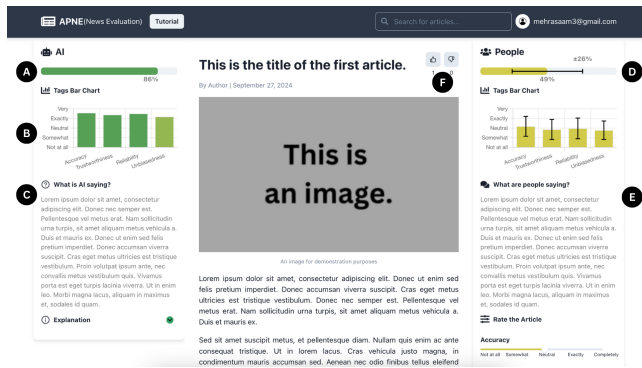


Figure 2: APNE Article Page. A) The AI rating score is shown in a color-coded bar. B) A bar chart of AI score for each criterion. C) The summary of AI’s reasoning. D) Other readers’ rating scores are shown in a color-coded bar along with the standard deviation. E) A teaser of other readers’ comments. F) An option to like or dislike the article. (Due to copyright restrictions, original images and text have been replaced with placeholders and Lorem Ipsum text.)

assessments while simultaneously considering both perspectives to make decisions.

3.2.1 AI Component. On the left side of the page is the AI Component, designed to provide users with an AI-generated assessment of the article. This component features an overall score, calculated based on four evaluation criteria — accuracy, reliability, trustworthiness, and unbiasedness. These criteria were inspired by prior works [15] which identified multiple factors contributing to credibility, such as accuracy, fairness, and public interest. Accuracy refers to the extent to which the news contains factual and precise information. Reliability assesses the credibility of the sources cited within the article. Trustworthiness evaluates whether the article is written in a manner that inspires confidence. Unbiasedness ensures that the author has not incorporated personal bias into the content.

The overall score is displayed using a horizontal, color-coded bar (Fig. 2 (A)), with individual criterion scores detailed in the accompanying AI Tag Chart positioned below (Fig. 2 (B)). Additionally, the AI provides a concise summary of its reasoning (Fig. 2 (C)). The summary can be expanded for a more detailed explanation to help users better understand the scoring process. The section can be minimized or maximized at any time to reduce distractions. The insights are generated using GPT-4o. In developing this component, we explored a range of models, from basic machine learning algorithms such as Random Forest [6] and SVM [24] to large language models (LLMs). Our objective was to identify a model capable of accurately detecting fake news while providing clear explanations to foster user understanding and trust—an essential factor, as users often struggle to trust AI without adequate reasoning. After evaluating various LLMs, we selected GPT-4o due to its performance in detecting fake news and its ability to generate clear and detailed explanations. We prompted the model with the following instruction: “Give the article a score from 0 to 100 for (1) accuracy, (2) trustworthiness, (3) reliability, and (4) unbiasedness. Then briefly

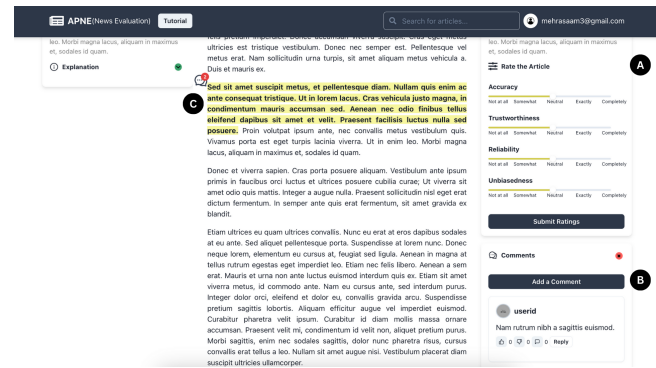


Figure 3: APNE Article Page with People’s Interactions. A) Rating option for each criterion. B) Adding a comment in the comment section. C) Selected sentence with a yellow highlight and an icon showing the number of comments for that sentence. (Due to copyright restrictions, original images and text have been replaced with placeholders and Lorem Ipsum text.)

explain your scores.” We also provided definitions of each assessment criterion to guide the explanation generation.

3.2.2 People Component. On the right side of the page, the People component is rendered. Similar to the AI component, this section features an overall score displayed at the top (Fig. 2 (D)). However, this score includes an additional standard deviation indicator, which highlights the variability in other readers’ evaluations. A larger standard deviation suggests significant differences in user opinions. Directly below the overall score, a histogram presents scores for each evaluation criterion, along with their respective standard deviations, offering a detailed breakdown of user ratings.

This component also includes a teaser summary of user discussions from the comment section (Fig. 2 (D)). To generate this summary, we used GPT-4o. Using GPT-4o, we processed all comments and their replies, prompting it to summarize the key points. Below the teaser, readers can provide their own ratings for the article based on predefined criteria (Fig. 3 (A)). The rating input is displayed as a set of color-coded sliders, allowing users to assign scores for each criterion. These ratings are dynamically reflected in the overall score and the histogram.

Readers can add comments by selecting a specific sentence in the Article Component. Once a sentence is clicked, the comment section automatically appears in the People Component. Above the comment section, there is an “Add a Comment” button (Fig. 3 (B)) that allows readers to leave a comment, with the option to include a source link for additional context. Users can engage with the discussion by liking or disliking comments or replying to them. To handle large volumes of comments effectively, the interface supports collapsing and expanding comment threads.

3.2.3 Article Component. The Article component presents the article’s content, including the image, title, author details, and main body of text. Additionally, this component allows readers to like or dislike the entire article (Fig. 2 (F)), providing another way to express their opinions. Readers can interact with this component

by hovering over individual sentences. When a sentence is hovered over, it becomes highlighted in blue, and an icon appears, indicating the number of comments associated with that specific sentence.

Readers can leave a comment by clicking on a highlighted sentence, which activates the comment section in the People component. At the bottom of the Article Component, a list of all comments is displayed. Selecting a specific comment highlights the related sentence in the article with a yellow highlight (Fig. 3 (C)), enabling readers to easily view discussions connected to that sentence in the People component. This option facilitates a better understanding of the conversation surrounding specific parts of the article so that readers do not have to read through the complete article to identify what part of the article a particular comment is referring to. Readers also have the option to minimize the comment section for a more focused reading experience, reducing cognitive overload.

4 Evaluation

We conducted an initial study with 20 participants to evaluate how readers utilize APNE to identify misinformation in news articles. We asked participants to read three news articles using APNE and share their usage experience, how the AI and other readers' comments impacted their perception of the articles, and suggestions for improvements. Our analysis of participants' responses suggests that while AI was appreciated for providing factual information, some participants found the AI to be biased. We also found that while people have subjective opinions, they can provide facts and alternative sources that could better inform other readers. Finally, we surfaced how participants' opinions on the articles were influenced by factors beyond the article content including source, their mistrust of AI, and their judgment toward other readers.

4.1 Participants

We recruited 20 participants through Prolific from U.S. 12 participants used she/her pronouns and 8 participants used he/him pronouns. The average age of our participants was 40 ± 15.60 years, ranging from 19 to 82. 60% (12 participants) identified various news websites as their primary news sources. When asked about their experiences with misinformation in the news, 70% of participants (14 out of 20) reported having prior experiences of encountering misinformation. In response to how they typically handle misinformation, participants indicated a range of strategies. Six of these participants stated that when exposed to misinformation, they simply ignore it. Others reported that they cross-check the information by searching other sources or discussing it with others to verify its accuracy.

4.2 Procedure

We populated APNE with three news articles. The evaluation was conducted in October 2024. All three articles were contemporary and on politically charged topics. We selected these articles, due to them being heavily interacted with in X, a popular social media platform, with the most impressions and comments, at the time of our data collection. We also populated the comments section with comments from X. Based on our internal assignment of ratings,

we converted each comment to rating scores. We created a tutorial video to explain APNE's components and guide participants through each step of the study.

The participants started their session by watching the tutorial video to familiarize themselves with APNE. This was followed by a pre-study questionnaire where we asked participants about their typical interaction with news, particularly misinformation in news articles. Once the questionnaire was completed, participants were directed to APNE's homepage. They were instructed to read each news article and review both the AI and people-generated assessments. They were then asked to rate each article based on the specified criteria and provide comments on any specific parts they found relevant. Participants spent an average of 22.7 ± 16.01 minutes reading and assessing the articles. After assessing all articles, participants completed a post-study questionnaire. This questionnaire aimed to gather their experiences of using APNE and feedback and opinions about the system's usability and effectiveness. Each participant was compensated at a rate of \$15.60 per hour.

We stored the pre- and post-study questionnaire alongside participants' interaction with APNE for analysis. We qualitatively analyzed the questionnaire responses using open coding [13]. In the following, we present our findings.

4.3 Findings

4.3.1 AI Provides Objective Assessments but can be Biased. Our participants preferred the AI component for its objective assessment, explanation of its reasoning, and highlighting facts. For instance, P18 mentioned, *"The AI-generated insights dealt directly with what the article said or didn't say/prove/show backup information for. The comments [from people] were mostly just off-the-cuff, not very thoughtful responses by mostly overly biased people."* However, some participants emphasized their distrust of the AI component, *"...I cannot overstate how much I distrust the ability of an AI to give an unbiased summary of the news, and I intrinsically distrust anything (human or machine) positioning itself as such an unbiased arbiter who will read the news so I don't have to."* (P20) Others mentioned that the AI itself is biased. For instance, P6 mentioned, *"While people are biased, at least you are going to get a mixed bag. This AI and the majority of all other AIs are biased, due to whomever programmed them and I have zero faith in their reliability or of being factual."* Interestingly, some participants directly opposed this perspective on AI bias, *"Unlike people, AI is unbiased and assesses the articles with the available resources and cross-checks it with other credible sources."* (P8) These findings suggest a divergent viewpoint towards AI's bias and how trustworthy they can be in providing factual information.

4.3.2 People's Subjective Opinions can be Both Thought-Provoking and Factual. Our participants emphasized the importance of seeing others' perceptions and how they can change the perspectives of the new articles. For instance, P9 mentioned, *"The other readers had interesting points that were not available in the article and were interesting to think about."* P7 explained how others' comments prompted them to reconsider their perspective on the article, *"There were comments that made me contemplate the article more than I would have."* Beyond diverse perspectives, some participants

highlighted how comments from other readers helped them identify biases in the article, *“In the Melania Trump article, someone’s comments I read pinpointed biases from the reporting source.”* (P3) P15 further added how others’ comments contradicted the AI’s assessment by validating events, *“About the attempt on Trump’s life, the people verified that that did happen, despite what the AI said.”* These findings paint a picture of how others’ comments can include factual information surpassing subjective opinions to influence the perspectives on news articles.

4.3.3 External Factors Influence People’s Perception of Online News Articles. From our participants’ responses, we identified external factors that influenced how they viewed articles. For instance, P20 declared not sufficiently engaging with the AI component, *“I didn’t significantly engage with the AI’s assessment of the article. ... I have no interest in having an AI assess news to add yet ‘another’ layer of bias and misdirection.”* In contrast, P3 mentioned preferring AI’s assessment of the article over other readers because *“... my [P3’s] thoughts aligned with AI analysis.”*

Many participants emphasized the importance of the source of the article with P10 saying, *“This article seemed highly unlikely. None of the writing in the letter sounded like anything Melania would say and there were no sources to back any of the information up.”* P20 shared similar sentiments about the same article but questioned the very source of the letter, *“... I would have liked to see some analysis or fact-checking from the reporters however, and I do not trust *her* [Melania Trump] words at all.”* Some participants completely dismissed others’ opinions; with P17 saying, *“The comments I read were more off-the-cuff judgments obviously made by very biased people. I read them because I find them amusing, mostly.”* P14 explained why others’ comments did not impact their decision, *“I don’t trust anyone that much.”* These responses suggest that people’s perception of news articles can go beyond the content of the article itself and can be influenced by external factors including pre-existing alignments, trust in the source, and technology used to explore news.

4.3.4 Participant Suggestions and Areas of Improvements. Our participants were generally positive about APNE, with 75% (15 participants) mentioning that they would use similar systems in the future if it was available. For instance, P9 mentioned, *“I’d like to see it [APNE] being used in real life.”* However, they also provided valuable suggestions for improvements. Many participants suggested adding external citations and specific references that could provide further support and increase trust. P12 mentioned, *“Adding direct citations to the sentences that could have misinformation or show bias... could further improve the credibility of the information.”* Others suggested adding *“...more detailed, nuanced feedback on the tone, style, and potential emotional influence of the article, in addition to focusing on factual accuracy and source verification.”* (P19) Several participants pointed out that it would be beneficial to add options to directly ask questions to the AI and receive more concise information. For instance, P10 mentioned, *“It might be helpful if you could ask the tool questions about the article directly in a chat box.”* Finally, some participants suggested potential improvements to the AI and interface, including making all components collapsible to reduce clutter (P20), making the AI system more unbiased (P6), and reducing the AI reasoning texts (P7).

5 Conclusion and Future Works

In this work, we designed and developed AI and People News Evaluation (APNE), a hybrid system leveraging AI and crowdsourcing to address misinformation. APNE combines AI’s objectivity with human insight to help users form informed opinions on news articles. Our initial evaluation of the tool with 20 participants revealed that while AI contributes factual insights, it may propagate bias, and human input can offer alternative perspectives for more informed decision-making. We also found that our participants’ opinions were influenced by factors such as source credibility, AI mistrust, and pre-existing beliefs.

In the future, we plan to further improve APNE by expanding the AI capability in two key areas. First, we will further explore large language models that can generate more accurate and reliable representations that can generate adjustable summaries [14]. We will also integrate vision language models (VLM) [29, 30] alongside LLMs to extract and identify information from pictures related to the articles. Second, we will extract and provide citations to external references to substantiate the reasoning and information generated by the AI component. To improve the People component, we will include additional visual cues that provide more background and information about other readers to build more credibility of the comments made by others. Finally, we will conduct longitudinal studies by deploying APNE in real-world settings and allow exploration of publicly available news articles to investigate its impact on identifying misinformation at a broader scale.

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